

Insurance

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Load Libraries and Data

```
# Load necessary libraries
library(car)
library(caret)
library(dplyr)
library(MASS)
# Load the data set
data <- read.csv("insurance.csv")
```

Encode Categorical Variables

```
# Encode the data
# Create pr_sex variable: 0 for female, 1 for male
data$pr_sex <- ifelse(data$sex == "male", 1, 0)

# Create pr_smoker variable: 0 for non-smoker, 1 for smoker
data$pr_smoker <- ifelse(data$smoker == "yes", 1, 0)

# Create dummy variables for region, excluding 'northwest' as the reference group
data$pr_region_northeast <- ifelse(data$region == "northeast", 1, 0)
data$pr_region_southeast <- ifelse(data$region == "southeast", 1, 0)
data$pr_region_southwest <- ifelse(data$region == "southwest", 1, 0)
```

Handle data

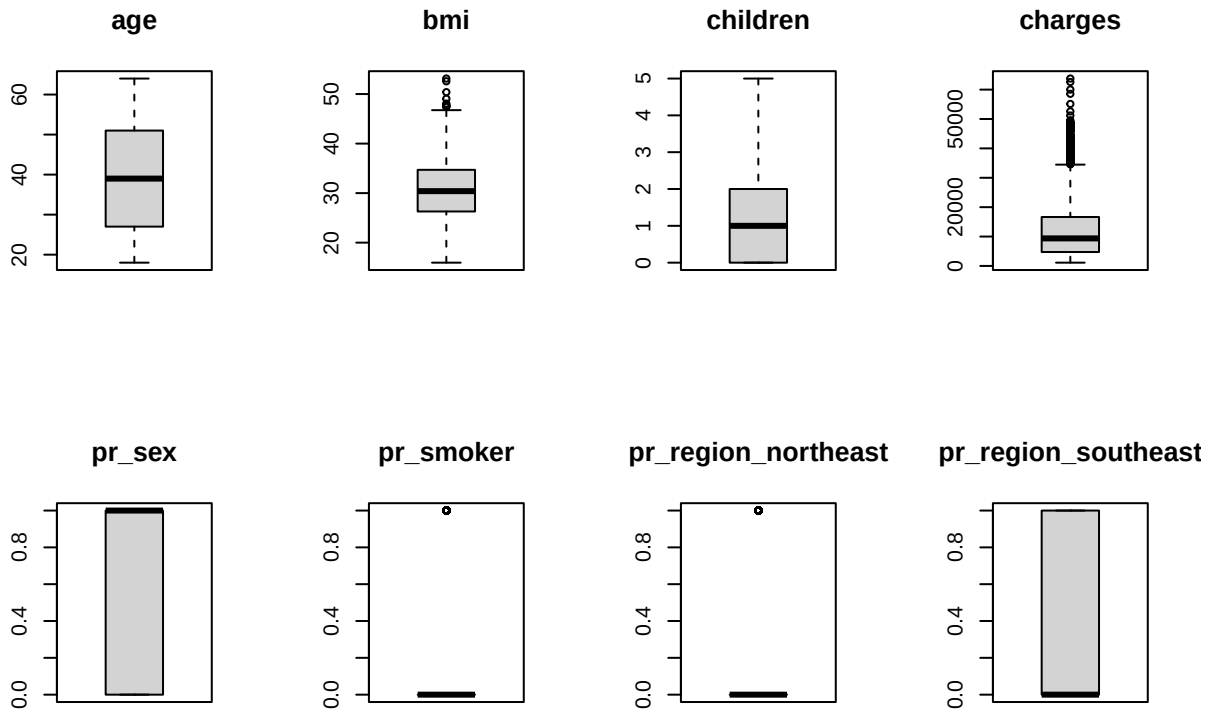
```
# Fit the initial model
initial_model <- lm(charges ~ pr_sex + pr_smoker + pr_region_northeast + pr_region_southeast + pr_region_southwest)

# Check for normality of residuals using Shapiro-Wilk test
shapiro_test <- shapiro.test(residuals(initial_model))
print(shapiro_test)
```

```
##
##  Shapiro-Wilk normality test
##
```

```
## data: residuals(initial_model)
## W = 0.89894, p-value < 2.2e-16
```

```
# Draw boxplots for all numeric columns to visualize outliers
numeric_columns <- sapply(data, is.numeric)
par(mfrow=c(2,4))
for(i in which(numeric_columns)){
  boxplot(data[,i], main=names(data)[i])
}
```



```
# Calculate Cook's distance
cooks_d <- cooks.distance(initial_model)

# Identify influential points (top 10% by Cook's distance)
influential <- as.numeric(names(sort(cooks_d, decreasing = TRUE)[1:floor(0.1 * length(cooks_d))]))

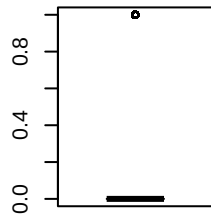
# Remove influential observations
data_clean <- data[-influential, ]

# Check for NA values and remove them
if(any(is.na(data_clean))){
  data_clean <- na.omit(data_clean)
}

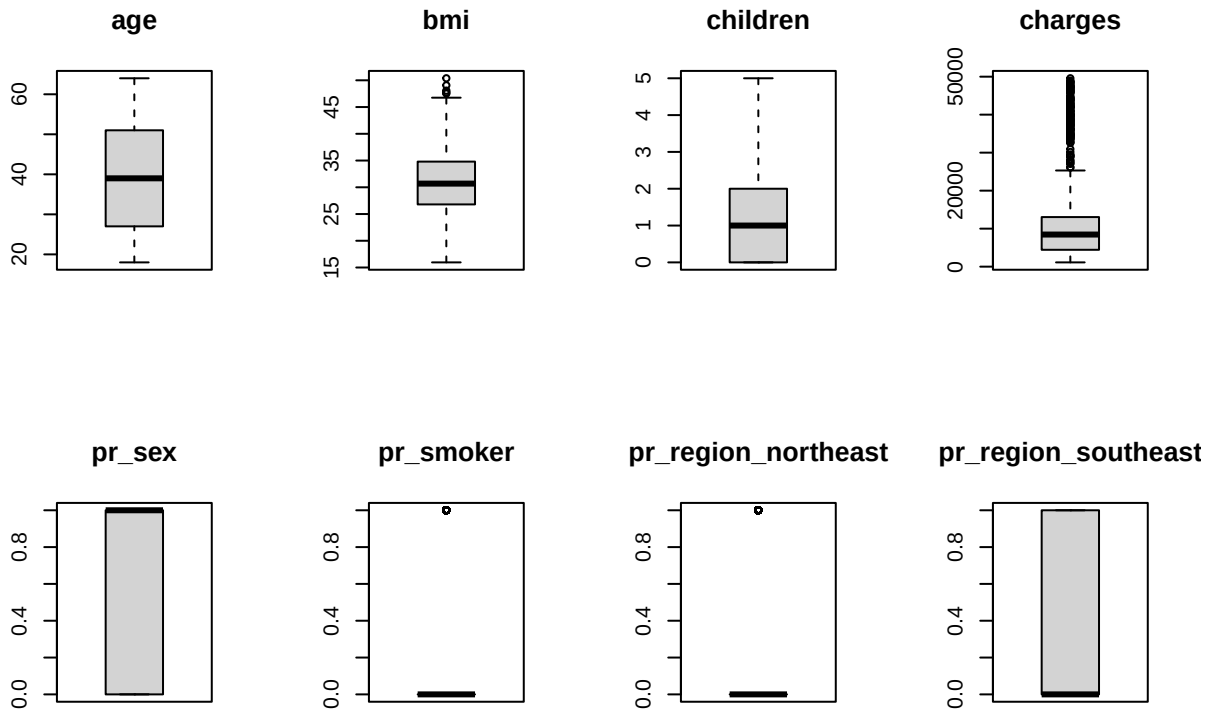
# Draw boxplots for all numeric columns to visualize outliers after cleaning
```

```
numeric_columns <- sapply(data_clean, is.numeric)
par(mfrow=c(2,4))
```

pr_region_southwest



```
for(i in which(numeric_columns)){
  boxplot(data_clean[,i], main=names(data_clean)[i])
}
```



```
# Percentage of data removed
```

```
(dim(data)[1] - dim(data_clean)[1]) / dim(data)[1] * 100
```

```
## [1] 9.940209
```

```
# Fit a linear model on the cleaned data to check for normality again
```

```
clean_model <- lm(charges ~ pr_sex + pr_smoker + pr_region_northeast + pr_region_southeast + pr_region_
```

```
shapiro_test_cleaned <- shapiro.test(residuals(clean_model))
```

```
print(shapiro_test_cleaned)
```

```
##
```

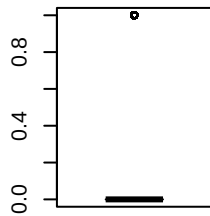
```
## Shapiro-Wilk normality test
```

```
##
```

```
## data: residuals(clean_model)
```

```
## W = 0.87312, p-value < 2.2e-16
```

pr_region_southwest



Calculate VIF

Calculate VIF

```
vif_data <- vif(lm(charges ~ pr_sex + pr_smoker + pr_region_northeast + pr_region_southeast + pr_region_southwest))
print(vif_data)
```

##	pr_sex	pr_smoker	pr_region_northeast	pr_region_southeast
##	1.012839	1.021390	1.509554	1.678562
##	pr_region_southwest	age	bmi	children
##	1.536423	1.021840	1.129036	1.005614

The VIF table indicates that the VIF values of six predictors are nearly close to 1, which means there is no severe multicollinearity problem among this model.

Split Data into Training and Testing Sets

Set seed for reproducibility
`set.seed(123)`

Split data

```
train_index <- createDataPartition(data_clean$charges, p = 0.8, list = FALSE)
train_data <- data_clean[train_index, ]
test_data <- data_clean[-train_index, ]
```

Train Linear Regression Model

Define Full and Initial Models

```
# Define the full model
modFull <- lm(charges ~ pr_sex + pr_smoker + pr_region_northeast + pr_region_southeast + pr_region_southwest, data = train_data)

# Define the null model
modZero <- lm(charges ~ 1, data = train_data)

# Define an intermediate model (you can customize the variables as needed)
modInter <- lm(charges ~ pr_sex + pr_smoker + pr_region_northeast + age + bmi, data = train_data)
```

Apply Step-Wise Method

```
# Backward elimination with AIC
modBest1A <- stepAIC(modFull, direction = "backward", scope = list(lower = modZero, upper = modFull), k = 1)

## Start:  AIC=16018.15
## charges ~ pr_sex + pr_smoker + pr_region_northeast + pr_region_southeast +
##         pr_region_southwest + age + bmi + children
##
##              Df  Sum of Sq      RSS   AIC
## - pr_region_southwest  1  1.4507e+06  1.5324e+10 16016
## - pr_sex               1  2.2394e+06  1.5325e+10 16016
## - pr_region_southeast  1  1.4864e+07  1.5337e+10 16017
## <none>                  1.5323e+10 16018
## - pr_region_northeast  1  1.2065e+08  1.5443e+10 16024
## - children             1  3.6621e+08  1.5689e+10 16039
## - bmi                  1  1.3684e+09  1.6691e+10 16099
## - age                  1  1.1801e+10  2.7123e+10 16567
## - pr_smoker            1  8.7864e+10  1.0319e+11 17857
##
## Step:  AIC=16016.24
## charges ~ pr_sex + pr_smoker + pr_region_northeast + pr_region_southeast +
##         age + bmi + children
##
##              Df  Sum of Sq      RSS   AIC
## - pr_sex               1  2.1962e+06  1.5326e+10 16014
## - pr_region_southeast  1  2.7079e+07  1.5351e+10 16016
## <none>                  1.5324e+10 16016
## - pr_region_northeast  1  1.4296e+08  1.5467e+10 16023
## - children             1  3.6584e+08  1.5690e+10 16037
## - bmi                  1  1.3886e+09  1.6713e+10 16098
## - age                  1  1.1799e+10  2.7123e+10 16565
## - pr_smoker            1  8.7875e+10  1.0320e+11 17855
##
## Step:  AIC=16014.37
## charges ~ pr_smoker + pr_region_northeast + pr_region_southeast +
##         age + bmi + children
##
```

```
##           Df Sum of Sq      RSS   AIC
## - pr_region_southeast 1 2.7076e+07 1.5353e+10 16014
## <none>                  1.5326e+10 16014
## - pr_region_northeast 1 1.4239e+08 1.5469e+10 16021
## - children             1 3.6506e+08 1.5691e+10 16035
## - bmi                  1 1.3866e+09 1.6713e+10 16096
## - age                  1 1.1805e+10 2.7131e+10 16564
## - pr_smoker            1 8.8255e+10 1.0358e+11 17856
##
## Step: AIC=16014.08
## charges ~ pr_smoker + pr_region_northeast + age + bmi + children
##
##           Df Sum of Sq      RSS   AIC
## <none>                  1.5353e+10 16014
## - pr_region_northeast 1 2.0233e+08 1.5556e+10 16025
## - children             1 3.7432e+08 1.5728e+10 16035
## - bmi                  1 1.3780e+09 1.6731e+10 16095
## - age                  1 1.1902e+10 2.7255e+10 16566
## - pr_smoker            1 8.8412e+10 1.0377e+11 17856
```

Backward elimination with BIC

```
modBest1B <- stepAIC(modFull, direction = "backward", scope = list(lower = modZero, upper = modFull), k
```

```
## Start: AIC=16064.94
## charges ~ pr_sex + pr_smoker + pr_region_northeast + pr_region_southeast +
##           pr_region_southwest + age + bmi + children
##
##           Df Sum of Sq      RSS   AIC
## - pr_region_southwest 1 1.4507e+06 1.5324e+10 16058
## - pr_sex               1 2.2394e+06 1.5325e+10 16058
## - pr_region_southeast 1 1.4864e+07 1.5337e+10 16059
## <none>                  1.5323e+10 16065
## - pr_region_northeast 1 1.2065e+08 1.5443e+10 16065
## - children             1 3.6621e+08 1.5689e+10 16080
## - bmi                  1 1.3684e+09 1.6691e+10 16140
## - age                  1 1.1801e+10 2.7123e+10 16609
## - pr_smoker            1 8.7864e+10 1.0319e+11 17898
##
## Step: AIC=16057.83
## charges ~ pr_sex + pr_smoker + pr_region_northeast + pr_region_southeast +
##           age + bmi + children
##
##           Df Sum of Sq      RSS   AIC
## - pr_sex               1 2.1962e+06 1.5326e+10 16051
## - pr_region_southeast 1 2.7079e+07 1.5351e+10 16052
## <none>                  1.5324e+10 16058
## - pr_region_northeast 1 1.4296e+08 1.5467e+10 16060
## - children             1 3.6584e+08 1.5690e+10 16073
## - bmi                  1 1.3886e+09 1.6713e+10 16134
## - age                  1 1.1799e+10 2.7123e+10 16602
## - pr_smoker            1 8.7875e+10 1.0320e+11 17891
##
## Step: AIC=16050.77
## charges ~ pr_smoker + pr_region_northeast + pr_region_southeast +
```

```
##      age + bmi + children
##
##              Df  Sum of Sq      RSS    AIC
## - pr_region_southeast  1 2.7076e+07 1.5353e+10 16045
## <none>                                1.5326e+10 16051
## - pr_region_northeast  1 1.4239e+08 1.5469e+10 16052
## - children              1 3.6506e+08 1.5691e+10 16066
## - bmi                   1 1.3866e+09 1.6713e+10 16127
## - age                   1 1.1805e+10 2.7131e+10 16595
## - pr_smoker             1 8.8255e+10 1.0358e+11 17888
##
## Step:  AIC=16045.27
## charges ~ pr_smoker + pr_region_northeast + age + bmi + children
##
##              Df  Sum of Sq      RSS    AIC
## <none>                                1.5353e+10 16045
## - pr_region_northeast  1 2.0233e+08 1.5556e+10 16051
## - children              1 3.7432e+08 1.5728e+10 16061
## - bmi                   1 1.3780e+09 1.6731e+10 16121
## - age                   1 1.1902e+10 2.7255e+10 16592
## - pr_smoker             1 8.8412e+10 1.0377e+11 17882
```

```
# Forward selection with BIC
```

```
modBest2 <- stepAIC(modZero, direction = "forward", scope = list(lower = modZero, upper = modFull), k =
```

```
## Start:  AIC=18001.22
## charges ~ 1
##
##              Df  Sum of Sq      RSS    AIC
## + pr_smoker      1 9.0295e+10 3.0671e+10 16684
## + age            1 1.2985e+10 1.0798e+11 17899
## + bmi            1 5.1144e+09 1.1585e+11 17967
## + children       1 1.1359e+09 1.1983e+11 17999
## <none>                                1.2097e+11 18001
## + pr_sex         1 5.1846e+08 1.2045e+11 18004
## + pr_region_southeast  1 4.7621e+08 1.2049e+11 18005
## + pr_region_southwest  1 2.7083e+07 1.2094e+11 18008
## + pr_region_northeast  1 1.1512e+07 1.2095e+11 18008
##
## Step:  AIC=16684.25
## charges ~ pr_smoker
##
##              Df  Sum of Sq      RSS    AIC
## + age            1 1.3443e+10 1.7228e+10 16135
## + bmi            1 2.4197e+09 2.8251e+10 16612
## + children       1 6.7351e+08 2.9997e+10 16670
## <none>                                3.0671e+10 16684
## + pr_region_northeast  1 1.8201e+08 3.0489e+10 16686
## + pr_region_southeast  1 1.3949e+07 3.0657e+10 16691
## + pr_region_southwest  1 2.2385e+05 3.0671e+10 16691
## + pr_sex         1 8.1377e+04 3.0671e+10 16691
##
## Step:  AIC=16134.83
## charges ~ pr_smoker + age
```

```
##
##              Df Sum of Sq      RSS   AIC
## + bmi              1 1304368609 1.5923e+10 16066
## + children          1  408246080 1.6819e+10 16119
## <none>                      1.7228e+10 16135
## + pr_region_northeast 1   82122867 1.7145e+10 16137
## + pr_region_southwest 1    936387 1.7227e+10 16142
## + pr_sex            1    752828 1.7227e+10 16142
## + pr_region_southeast 1    168297 1.7227e+10 16142
##
## Step:  AIC=16066.05
## charges ~ pr_smoker + age + bmi
##
##              Df Sum of Sq      RSS   AIC
## + children          1 367656815 1.5556e+10 16051
## + pr_region_northeast 1 195667220 1.5728e+10 16061
## <none>                      1.5923e+10 16066
## + pr_region_southeast 1 100628622 1.5823e+10 16067
## + pr_sex            1    870506 1.5922e+10 16073
## + pr_region_southwest 1    757778 1.5922e+10 16073
##
## Step:  AIC=16050.71
## charges ~ pr_smoker + age + bmi + children
##
##              Df Sum of Sq      RSS   AIC
## + pr_region_northeast 1 202330260 1.5353e+10 16045
## <none>                      1.5556e+10 16051
## + pr_region_southeast 1  87014088 1.5469e+10 16052
## + pr_sex            1   1493824 1.5554e+10 16058
## + pr_region_southwest 1   1476480 1.5554e+10 16058
##
## Step:  AIC=16045.27
## charges ~ pr_smoker + age + bmi + children + pr_region_northeast
##
##              Df Sum of Sq      RSS   AIC
## <none>                      1.5353e+10 16045
## + pr_region_southeast 1  27076363 1.5326e+10 16051
## + pr_region_southwest 1  13548685 1.5340e+10 16052
## + pr_sex            1   2193965 1.5351e+10 16052
```

```
# Stepwise selection with BIC
```

```
modBest3 <- stepAIC(modInter, direction = "both", scope = list(lower = modZero, upper = modFull), k = 1)
```

```
## Start:  AIC=16068.43
## charges ~ pr_sex + pr_smoker + pr_region_northeast + age + bmi
##
##              Df Sum of Sq      RSS   AIC
## + children          1 3.7511e+08 1.5351e+10 16052
## - pr_sex            1 1.4042e+06 1.5728e+10 16061
## <none>                      1.5726e+10 16068
## - pr_region_northeast 1 1.9620e+08 1.5922e+10 16073
## + pr_region_southeast 1 3.6344e+07 1.5690e+10 16073
## + pr_region_southwest 1 1.5830e+07 1.5710e+10 16075
## - bmi              1 1.4187e+09 1.7145e+10 16145
```

```
## - age          1 1.2122e+10 2.7848e+10 16613
## - pr_smoker    1 8.8333e+10 1.0406e+11 17885
##
## Step: AIC=16052.33
## charges ~ pr_sex + pr_smoker + pr_region_northeast + age + bmi +
##      children
##
##              Df Sum of Sq      RSS   AIC
## - pr_sex      1 2.1940e+06 1.5353e+10 16045
## <none>                                1.5351e+10 16052
## - pr_region_northeast 1 2.0303e+08 1.5554e+10 16058
## + pr_region_southeast 1 2.7079e+07 1.5324e+10 16058
## + pr_region_southwest 1 1.3665e+07 1.5337e+10 16059
## - children     1 3.7511e+08 1.5726e+10 16068
## - bmi          1 1.3800e+09 1.6731e+10 16128
## - age          1 1.1896e+10 2.7247e+10 16599
## - pr_smoker    1 8.8029e+10 1.0338e+11 17886
##
## Step: AIC=16045.27
## charges ~ pr_smoker + pr_region_northeast + age + bmi + children
##
##              Df Sum of Sq      RSS   AIC
## <none>                                1.5353e+10 16045
## - pr_region_northeast 1 2.0233e+08 1.5556e+10 16051
## + pr_region_southeast 1 2.7076e+07 1.5326e+10 16051
## + pr_region_southwest 1 1.3549e+07 1.5340e+10 16052
## + pr_sex          1 2.1940e+06 1.5351e+10 16052
## - children     1 3.7432e+08 1.5728e+10 16061
## - bmi          1 1.3780e+09 1.6731e+10 16121
## - age          1 1.1902e+10 2.7255e+10 16592
## - pr_smoker    1 8.8412e+10 1.0377e+11 17882
```

These 4 models are exactly one. So we choose the modBest1A. `### Summary of Best Model`

```
summary(modBest1A)
```

```
##
## Call:
## lm(formula = charges ~ pr_smoker + pr_region_northeast + age +
##      bmi + children, data = train_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12141.5  -1306.2   -215.4   1146.0  13794.2
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -9552.814     759.340  -12.580 < 2e-16 ***
## pr_smoker      25826.696     347.539   74.313 < 2e-16 ***
## pr_region_northeast 1085.813     305.432    3.555 0.000396 ***
## age           250.949        9.204   27.266 < 2e-16 ***
## bmi           206.229       22.229    9.278 < 2e-16 ***
## children      523.357       108.235    4.835 1.55e-06 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4001 on 959 degrees of freedom
## Multiple R-squared:  0.8731, Adjusted R-squared:  0.8724
## F-statistic: 1319 on 5 and 959 DF,  p-value: < 2.2e-16
```

Evaluate Model Performance

```
# Predict on test data
predictions <- predict(modBest1A, newdata = test_data)

# Calculate performance metrics
actual <- test_data$charges
mse <- mean((predictions - actual)^2)
rmse <- sqrt(mse)
r_squared <- cor(predictions, actual)^2

# MSE
mse
```

```
## [1] 14044199
```

```
# RMSE
rmse
```

```
## [1] 3747.559
```

```
# R-squared
r_squared
```

```
## [1] 0.9160806
```

Mean Squared Error (MSE): 14,044,199

Interpretation: MSE is the average of the squares of the errors between predicted and actual values. It gives a rough idea of how large the errors are. In this case, the MSE is quite large, indicating that there are significant differences between the predicted and actual values.

Root Mean Squared Error (RMSE): 3,747.559

Interpretation: RMSE is the square root of MSE and provides a measure of the average magnitude of the prediction errors in the same units as the response variable. An RMSE of 3,747.559 means that the typical error in predicting the charges is about 3,747.559 units.

R-squared (R^2): 0.9160806

Interpretation: R-squared represents the proportion of the variance in the dependent variable that is predictable from the independent variables. An R-squared value of 0.9160806 means that approximately 91.61% of the variability in the charges can be explained by the model. This is a high value, indicating that the model fits the data well.

Comments on Model Performance

High R-squared

The R-squared value of 0.9160806 suggests that the model explains a large portion of the variance in the response variable (charges). This implies that the predictors used in the model are quite effective in capturing the relationship between the inputs and the output.

High RMSE

While the R-squared value is high, the RMSE value of 3,747.559 indicates that there is still a considerable amount of error in the predictions. This suggests that while the model captures the overall trend well, individual predictions may still be off by a significant margin.

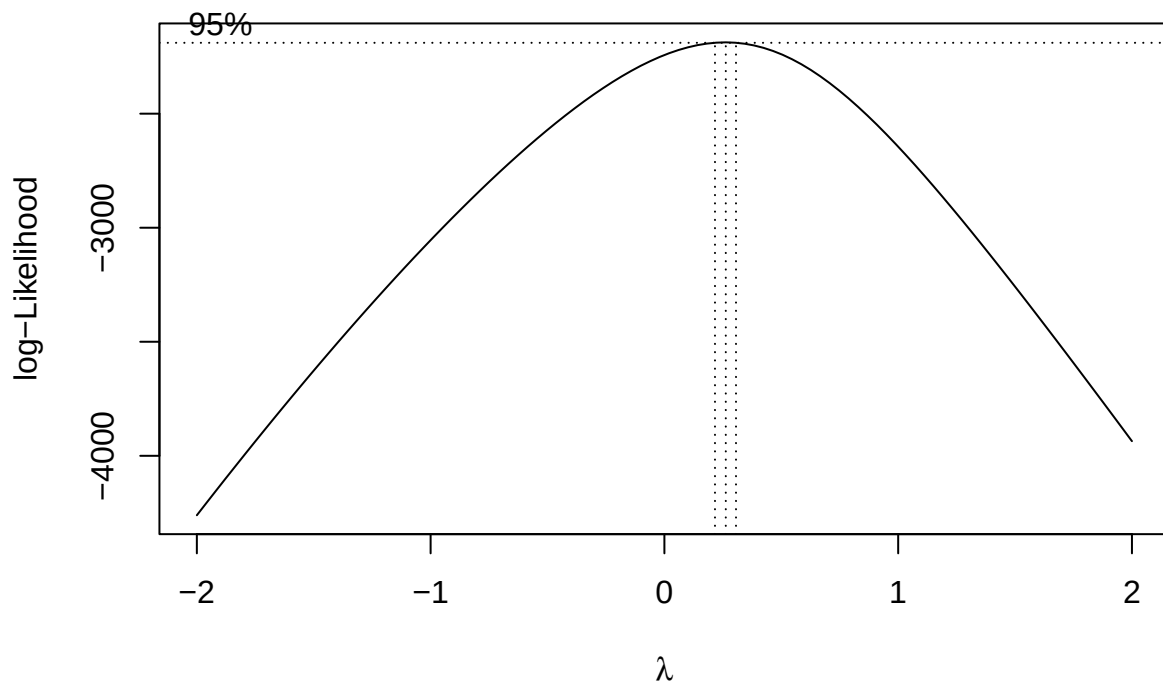
Model Appropriateness

Given the high R-squared value, the model seems to be a good fit for the data. However, the high RMSE indicates that there might be room for improvement, possibly by refining the model, adding more predictors, or considering non-linear relationships.

Apply Box-Cox transformation

```
# Fit a linear model on the training data
initial_model <- lm(charges ~ pr_sex + pr_smoker + pr_region_northeast + pr_region_southeast + pr_region_west)

# Find the optimal lambda for Box-Cox transformation
boxcox_result <- boxcox(initial_model, lambda = seq(-2, 2, by = 0.1))
```



```
lambda_optimal <- boxcox_result$x[which.max(boxcox_result$y)]

# Apply the Box-Cox transformation
if(lambda_optimal == 0) {
  train_data$charges_transformed <- log(train_data$charges)
  test_data$charges_transformed <- log(test_data$charges)
} else {
  train_data$charges_transformed <- (train_data$charges^lambda_optimal - 1) / lambda_optimal
  test_data$charges_transformed <- (test_data$charges^lambda_optimal - 1) / lambda_optimal
}

# Fit the model with the transformed response variable
modBestTransformed <- lm(charges_transformed ~ pr_sex + pr_smoker + pr_region_northeast + pr_region_southeast + pr_region_southwest + age + bmi + children,
  data = train_data)

# Summary of the transformed model
summary(modBestTransformed)
```

```
##
## Call:
## lm(formula = charges_transformed ~ pr_sex + pr_smoker + pr_region_northeast +
##     pr_region_southeast + pr_region_southwest + age + bmi + children,
##     data = train_data)
##
## Residuals:
```

##	Min	1Q	Median	3Q	Max
----	-----	----	--------	----	-----

```
## -10.3394 -1.3681 -0.1747 0.7547 16.6510
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    15.293742   0.640975  23.860 < 2e-16 ***
## pr_sex         -0.475115   0.213703  -2.223 0.026432 *
## pr_smoker      20.058696   0.288288  69.579 < 2e-16 ***
## pr_region_northeast 1.037239   0.305165   3.399 0.000704 ***
## pr_region_southeast -0.508572   0.310256  -1.639 0.101501
## pr_region_southwest -0.069832   0.301753  -0.231 0.817037
## age            0.357799   0.007612  47.004 < 2e-16 ***
## bmi            0.134701   0.019071   7.063 3.13e-12 ***
## children       0.886475   0.089471   9.908 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.304 on 956 degrees of freedom
## Multiple R-squared:  0.887, Adjusted R-squared:  0.8861
## F-statistic: 938 on 8 and 956 DF, p-value: < 2.2e-16
```

```
# Evaluate model performance on test data with transformed response variable
predictions_transformed <- predict(modBestTransformed, newdata = test_data)
actual_transformed <- test_data$charges_transformed
```

```
# Calculate performance metrics for transformed model
mse_transformed <- mean((predictions_transformed - actual_transformed)^2)
rmse_transformed <- sqrt(mse_transformed)
r_squared_transformed <- cor(predictions_transformed, actual_transformed)^2
```

```
# MSE for transformed model
mse_transformed
```

```
## [1] 8.393847
```

```
# RMSE for transformed model
rmse_transformed
```

```
## [1] 2.897214
```

```
# R-squared for transformed model
r_squared_transformed
```

```
## [1] 0.9232725
```